

Cross-View Gait Recognition by Discriminative Feature Learning

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Abstract—Recently, deep learning-based cross-view gait recognition has become popular owing to the strong capacity of convolutional neural networks (CNNs). Current deep learning methods often rely on loss functions used widely in the task of face recognition, e.g., contrastive loss and triplet loss. These loss functions have the problem of hard negative mining. In this paper, a robust, effective, and gait-related loss function, called angle center loss (ACL), is proposed to learn discriminative gait features. The proposed loss function is robust to different local parts and temporal window sizes. Different from center loss which learns a center for each identity, the proposed loss function learns multiple sub-centers for each angle of the same identity. Only the largest distance between the anchor feature and the corresponding cross-view sub-centers is penalized, which achieves better intra-subject compactness. We also propose to extract discriminative spatial-temporal features by local feature extractors and a temporal attention model. A simplified spatial transformer network is proposed to localize the suitable horizontal parts of the human body. Local gait features for each horizontal part are extracted and then concatenated as the descriptor. We introduce long short-term memory (LSTM) units as the temporal attention model to learn the attention score for each frame, e.g., focusing more on discriminative frames and less on frames with bad quality. The temporal attention model shows better performance than the temporal average pooling or gait energy images (GEI). By combining the three aspects, we achieve state-of-the-art results on several cross-view gait recognition benchmarks.

Index Terms—Gait recognition, discriminative feature learning, angle center loss, spatial-temporal features.

I. INTRODUCTION

GAIT is a kind of behavioral biometric feature, which is defined as the way a person walks. Compared with common biometric features such as face and iris, gait can be applied in long-distance and non-cooperative cases. One typical application of gait recognition is the suspect retrieval

in surveillance scenarios. Gait recognition has been used for evidence collection in Denmark [1] and UK [2].

Gait can be sensitive to some factors including view, clothing, walking speed, shoes and carrying status. Among them, the view change is a very tricky problem as appearance differences are introduced for different views. Gait energy images (GEI) [3] which are the average silhouettes along the temporal dimension have been widely used as the spatial-temporal gait features. The noise in silhouettes could be reduced while dynamic information may also be reduced at the same time. For the case without view changes, calculating L2 distance based on GEI achieves good results. Another easier case is multi-view gait recognition. It means that gait sequences from multiple views are provided in the gallery. The probe view could often find the corresponding view in the gallery, which reduces the difficulty of identification. However, it is hard to capture gait sequences in multiple views in real surveillance cases. Cross-view gait recognition, which only provides one gallery view, is necessary to alleviate the lack of registered views. Accordingly, the intra-subject variations are often larger than inter-subject variations, which makes the problem very difficult. We focus on cross-view gait recognition in this paper.

Approaches can be divided into two typical categories for cross-view gait recognition: model-based and appearance-based. Model-based methods try to reconstruct the 3D model of a person [4], [5]. This usually relies on multiple high resolution cameras. The imaging distance is also restricted, which limits its potential use. On the other hand, appearance-based methods extract features from raw videos from common surveillance cameras. This alleviates the need for high-end capture devices. However, models are often affected by appearance changes. Our method here belongs to appearance-based methods since we only rely on the sequences of binary gait silhouettes or GEI.

Appearance-based methods can be further divided into two categories: discriminative and generative approaches. Discriminative methods aim at directly learning view-invariant features from data. These methods include linear discriminant analysis (LDA) [6], Rank-SVM [7] and CNN methods [8]–[11]. Feature representation is very important for gait recognition. Previous non-deep learning methods often rely on hand-crafted features such as GEI [3] or chrono-gait images (CGI) [12]. A coupled patch alignment is one of the most effective approach in non-deep learning methods [13], which first builds a certain amount of patches, and each of them is made up of a sample as well as its intra-class and inter-class

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nearest-neighbors, then designs an objective function for each patch to balance the cross-view intra-class compactness and the cross-view inter-class separability, and finally, all the local independent patches are combined to render a unified objective function. Ben et al. also explored a general tensor representation framework for cross-view gait recognition [14]. Their approach is based on three criteria of tensorial coupled mappings, i.e. coupled multi-linear locality-preserved criterion, coupled multi-linear marginal fisher criterion, and coupled multi-linear discriminant analysis criterion. This approach is highly advantageous as it avoids the under-sample problem. They also analyze on the convergence for the first time [14].

Deep learning methods learn discriminative feature representations or metrics directly from raw data and have achieved promising results for gait recognition. Wu *et al.* [9] learned pair-wise similarities by a 3D CNN with temporal input. Takemura *et al.* [11] further improved the pair-wise similarity learning methods of [9] by optimizing the input, output and loss functions of CNN-based methods on large scale cross-view datasets. GEINet [10] used a multi-class classification method to learn gait features from GEI with softmax loss. Generative approaches usually transform gait features from different conditions into the same condition for better matching. Yu *et al.* [15] constructed an encoder-decoder network to transform the GEI images to the common condition. Generative adversarial networks (GAN) [16] were also introduced to convert an arbitrary gait condition into a side-view condition with normal clothing [17]. He *et al.* [18] proposed multi-task generative adversarial networks (MGANs) for learning view-specific feature representations. However, generative methods are not necessarily optimal in the sense of accuracy because they essentially aim at the synthesis of gait features among different conditions [11].

Deep learning based methods suffer from several problems when applied for cross-view gait recognition.

- (1) The loss functions used widely in face recognition cannot guarantee satisfying performance for gait recognition. Existing approaches [9]–[11] used softmax loss, contrastive loss or triplet loss for cross-view gait recognition. These loss functions have either discrimination or sampling problems. We find that there are seldom gait-specific loss functions. The view information is often neglected during the design of loss functions. The exception is that of [18] which learned view-specific feature representations.
- (2) The spatial-temporal feature learning could not capture detailed gait features. Most existing approaches rely on the whole body rather than the local parts as input. Although [19] and [20] considered part selections, these methods often neglect view changes. The discriminative body parts would be different for cross-view gait recognition. Approaches have also been proposed to capture temporal gait features. 3D convolution was used by [21], [22] to model dynamic features of gait sequences. Long-short term memory (LSTM) units were used in [23] and [24] to model the movement of human keypoints. However dynamic gait features may be lost due to the short time window used by these methods.

To overcome these problems, we could borrow ideas from face recognition, person re-identification and action recognition. In face recognition, center loss [25] and feature normalization based methods [26]–[28] have been proposed to learn better discriminative features. In person re-identification, spatial feature learning methods including spatial saliency matching [29], part-based learning [30], [31], and mask-guided learning [32] have been proved to be effective. These approaches force the model to focus on the discriminative parts rather than the background or the global human body. For action recognition, temporal modeling such as C3D [33], LSTM [34] and two-stream networks [35] could be used.

Accordingly, we focus on discriminative loss functions and spatial-temporal feature learning for cross-view gait recognition.

- (1) Inspired by the loss functions in face recognition, a robust, effective and gait-related loss function, called angle center loss, is proposed with the following advantages.

Robust: The proposed loss function has only one major hyper-parameter to control the intra-subject compactness which alleviates the need for the hyper-parameter grid search. It is robust to different local parts and temporal window sizes in gait recognition. It also avoids the sampling problem of contrastive loss and triplet loss.

Effective: Several state-of-the-art loss functions are carefully evaluated, and we find that the proposed loss function performs well for both small and large datasets.

Gait-related: Sub-centers are learned for each angle of the same identity. The update strategy of centers follows the standard back propagation which is efficient at training time. Only the largest distance between the anchor feature and the corresponding cross-view sub-centers is penalized, which matches the cross-view gait recognition setting.

- (2) We propose an effective spatial-temporal feature learning approach. For spatial feature learning, we find that local features with the learned partition achieve better performance compared with global features and uniform local features. By analyzing the results of different body parts, we conclude that head and upper legs are important for cross-view gait recognition. For temporal feature learning, we propose an LSTM attention model to learn the attention weight for each frame. The spatial-temporal features are combined as the final feature representation for a gait sequence. Cosine similarities are computed on these feature vectors.

We summarize our main contributions as follows.

- For the problem of low discrimination in softmax loss and hard negative mining in contrastive/triplet loss, we propose a robust, effective and gait-related loss function for cross-view gait recognition. We also compare different loss functions for gait recognition.
- For the problem of coarse spatial-temporal feature extraction, we learn effective spatial-temporal features to further boost performance by a learned horizontal partition and an LSTM attention model.

- For the problem of relatively low gait recognition accuracy, we combine the above aspects and achieve the state-of-the-art performance on several benchmarks including CASIA-B, OULP and OUMVLP.

In the remaining part of this paper, Section II summarizes related work on cross-view gait recognition. Section III describes the proposed method in detail. A comprehensive study on several benchmarks is given in Section IV. The conclusion of this paper is given in Section V.

II. RELATED WORK

A. Gait Recognition

Gait recognition approaches can be grouped into two major categories: model-based and appearance-based. In [4] and [5], the 3D structure of a person was reconstructed to generate arbitrary 2D views from the 3D model. These methods often require multiple calibrated cameras under fully-controlled and cooperative environments. With the success of 2D pose estimation [36] and 3D pose estimation [37] algorithms, model-based methods could be improved.

Appearance-based methods, on the other hand, extract gait representations directly from data. The accuracy of these methods is influenced greatly by view or clothing changes. Among several classic methods, view transformation model (VTM) is the most commonly used model. Gait features are transformed from one view to another in [38] and [39]. However, these methods rely on the provided view information. As for view invariant features, Hu *et al.* [40] proposed a method, named ViDP, to extract view invariant features using a unitary linear transform. In [41] and [42], hand-crafted view-invariant features were extracted for gait recognition.

Recently a lot of deep learning based methods have been proposed.

- **Generative methods.** Yu *et al.* [15] proposed a uni-form model for cross-view gait recognition with an encoder-decoder framework. Shiraga *et al.* [10] proposed a multi-class classification framework to learn view invariant features. In [17], generative adversarial networks (GAN) were used to generate invariant side-view gait images in normal clothing. Yu *et al.* [43] further improved the GAN-based method by adopting a multi-loss strategy to optimize the network to increase the inter-class distance and to reduce the intra-class distance at the same time. He *et al.* [18] proposed multi-task generative adversarial networks (MGANs) for learning view-specific feature representations. Hu *et al.* [44] proposed a Discriminant Gait Generative Adversarial Network (DiGGAN) framework to learn view-invariant gait features for cross-view gait recognition. The method could also reconstruct the gait templates in all views, which can be served as important evidences for forensic applications.
- **Discriminative methods.** Wu *et al.* [8] regarded gait recognition as an image set classification problem. Pose-based gait recognition was proposed by [24] to extract robust features against clothing and carrying changes. Wu *et al.* [9] proposed deep neural networks

to learn similarities for each pair. The pair-wise similarity learning method introduces much more samples and would alleviate overfitting. To merge information from the pairs, different levels of information fusion schemes were proposed. And for temporal features, a 3D convolutional neural network was proposed. In [11], cross-view gait recognition was addressed in terms of the input/output architecture, that is, the contrastive loss/triplet ranking loss for verification/identification and the high/low-level difference structure for large/small view difference. Chao *et al.* [45] proposed a new network named GaitSet to learn identity information from a set of silhouettes and achieved promising results.

- **Temporal modeling.** Temporal modeling is an important cue in gait recognition. Previous methods follow the standard pipeline of “CNN + LSTM” which is similar to action recognition [46]. Feng *et al.* [23] extracted heat maps by a CNN-based pose estimation method and then model the gait sequence by an LSTM recurrent neural network. Liao *et al.* [24] proposed a pose-based temporal-spatial network to extract the spatial-temporal features from gait pose rather than image. An *et al.* [47] used 3D pose estimation and achieved better performance compared with 2D pose estimation. Batchuluun *et al.* [48] used shallow CNN stacked with LSTM and deep CNN followed by score fusion to capture more spatial and temporal features. Battistone and Petrosino [49] proposed Time based Graph Long Short-Term Memory (TGLSTM) model to learn long short-term dependencies together with graph structure. Zhang *et al.* [50] proposed a novel auto-encoder framework to explicitly disentangle pose and appearance features from RGB imagery and the LSTM-based integration of pose features over time.

B. Discriminative Feature Learning

Recently face recognition, person re-identification and action recognition have achieved great improvements due to discriminative loss functions and spatial-temporal feature learning. We could borrow ideas from these fields.

- (1) **Loss functions.** Multi-class classification softmax loss was widely used in face recognition [51]. However, the learned features are not discriminative enough. Pair-wise contrastive loss [52] or triplet loss [53] were proposed with the problem of hard negative mining. To solve these problems, Wen *et al.* [25] proposed center loss to push the features to the corresponding class centers. Large margin softmax (LSoftmax) [54] is the first method to introduce angular constraints for each identity to improve feature discrimination. An annealing strategy has to be used to set the hyper-parameter to avoid the network divergence. Feature normalization based loss functions, including COCO loss [26], CosFace [27] and ArcFace [28], have also been proposed by modifying the original form of softmax loss for discriminative feature learning. It should be noted that current loss functions for face recognition are not designed for specific tasks, e.g., identification or verification. It has been proved that

the discriminative feature embedding can handle both identification and verification in [53].

- (2) **Spatial-temporal feature learning.** In [29], person re-identification was solved by proposing a salience matching strategy. Yi *et al.* [30] proposed a method that jointly learns the color feature, texture feature and metric in a unified framework. Li *et al.* [55] learned and localized deformable pedestrian parts using spatial transformer networks (STN) [56] with spatial constraints. Song *et al.* [32] took human segmentation as a mask to guide the network to focus on the human body. Sun *et al.* [31] achieved promising results with a concatenated descriptor consisting of several part-level local features. In action recognition, methods have been proposed to capture dynamic features such as [34], [57]. Several approaches to average features across the temporal dimension have been discussed in [58]. Chen *et al.* [59] proposed a part-activated deep reinforcement learning (PA-DRL) method to exploit the structure of the human body. This method captures the structural information of human body to describe actions. Tang *et al.* [60] proposed a progressive process through deep reinforcement learning to distil the most informative frames and discard ambiguous frames. After capturing the key frames, the graph-based convolutional neural network is proposed to capture the dependency between the joints for action recognition.

III. PROPOSED METHOD

A. Overall Framework

Figure 1 shows the overall framework. The proposed method can extract spatial-temporal features with the supervision of the gait-related loss function. The silhouette is divided into 4 horizontal parts by a learned partition. Each horizontal part is fed into an individual CNN. The LSTM attention model is used to output attention scores on the frame-level CNN. The frame-level features are averaged by the attention weight for each part. At the training stage, these different weighted features are fed into separate loss functions. At the testing stage, the weighted features for each part are concatenated together as a feature vector. Cosine similarities are calculated between these feature vectors for both the verification and the identification tasks.

B. Review of Loss Functions

We first review several loss functions used in face recognition.

The classic loss function is softmax loss with the formulation as follows:

$$L_s = - \sum_{i=1}^m \log \frac{e^{\mathbf{W}_{y_i}^T \mathbf{x}_i + b_{y_i}}}{\sum_{j=1}^n e^{\mathbf{W}_j^T \mathbf{x}_i + b_j}} \quad (1)$$

where $\mathbf{x}_i \in \mathbb{R}^d$ represents the i -th deep feature, belonging to the y_i -th class. d is the feature dimension. $\mathbf{W}_j \in \mathbb{R}^d$ denotes the j -th column of the weights $\mathbf{W} \in \mathbb{R}^{d \times n}$ in the last fully

connected layer and $\mathbf{b} \in \mathbb{R}^n$ is the bias term. The batch size and the class number are m and n respectively.

In face recognition, it has been proved that the learned features by the softmax loss are only separable rather than discriminative [25]. Center loss is thus proposed to learn discriminative features with the following formulation:

$$L_c = \frac{1}{2} \sum_{i=1}^m \|\mathbf{x}_i - \mathbf{c}_{y_i}\|_2^2 \quad (2)$$

where $\mathbf{c}_{y_i}$ represents the y_i th class center. Center loss penalizes the distances between the anchor feature and its corresponding class centers. The update of $\mathbf{c}_{y_i}$ is computed as:

$$\Delta \mathbf{c}_j = \frac{\sum_{i=1}^m \delta(y_i = j)(\mathbf{c}_j - \mathbf{x}_i)}{1 + \sum_{i=1}^m \delta(y_i = j)} \quad (3)$$

$$\mathbf{c}_j^{t+1} = \mathbf{c}_j^t - \alpha \Delta \mathbf{c}_j^t \quad (4)$$

where $\delta(\text{condition}) = 1$ if the condition is satisfied, and $\delta(\text{condition}) = 0$ if not. To avoid large perturbations caused by few mislabelled samples, α is introduced to control the learning rate of the centers.

Using only the center loss is not enough since the centers will barely change according to its formulation and thus the loss could be zero. The center loss should be used with the softmax loss to form the objective function as follows:

$$L = L_s + \lambda L_c \quad (5)$$

where the scalar λ is used to balance the two loss functions. By the softmax loss, the features are separable. And with the center loss, the features become discriminative.

C. Weight Normalization and Feature Normalization

Recent studies [26]–[28] have shown the effectiveness of feature normalization and weight normalization. By fixing the bias term $b_j = 0$, the target logit [61] can be rewritten as follows:

$$\mathbf{W}_j^T \mathbf{x}_i = \|\mathbf{W}_j\| \|\mathbf{x}_i\| \cos \theta_j \quad (6)$$

To remove the effect of $\mathbf{W}$, we fix $\|\mathbf{W}_j\| = 1$. As pointed out above, the similarities are usually calculated as the cosine similarity which is not related to the L2-norm of the features. The feature vectors are then fixed as $\|\mathbf{x}\| = s$. Feature and weight normalization will remove the radial variation and push every feature to distribute on a hypersphere manifold. However, the scale factor s is hard to select. Although a lower bound can be given in [26] and [27], the hyper-parameter is often set empirically for face recognition.

D. Angle Center Loss

In center loss, the center of each class is updated in a statistical manner as shown in Equation 3. We note that the gradient of center loss with respect to $\mathbf{c}_j$ can be calculated by current automatic differentiation frameworks such as MXNet [62]. And thus the update strategy can be simplified as the standard mini-batch stochastic gradient descent (SGD) algorithm:

$$\mathbf{c}_j^{t+1} = \mathbf{c}_j^t - \alpha \frac{\partial L_c}{\partial \mathbf{c}_j} \quad (7)$$

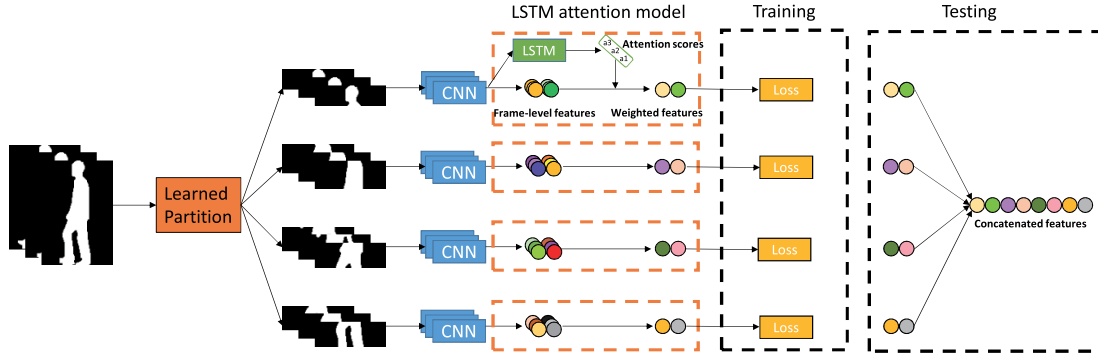


Fig. 1. The overall framework of our method. Angle center loss is proposed to learn discriminative features. The learned partition and the LSTM temporal attention model are used to extract spatial-temporal gait features.

The difference could be the learning rate and the weight decay of the centers compared with the original center loss. The learning rate should be fixed as α and the weight decay should be disabled for the center parameters to cope with the original form of center loss. With these modifications, we could achieve better efficiency and similar performance to the original center loss. Such efficiency enables us for the cases with more centers, e.g., sub-centers as described in the following.

For gait recognition, the center loss treats each view of an identity equally. In other words, all views from the same identity are pushed to the same center. In this paper, we introduce sub-centers for each view of the same identity. The sub-center for the y_i th class and the a_k th angle could be written as $\mathbf{c}_{y_ia_k}$. In this way, the total number of centers is enlarged as:

$$N_{sub} = N_v * N_c \quad (8)$$

where N_{sub} represents the number of all sub-centers, N_v the number of views and N_c the number of all identities.

We propose a hard positive mining strategy for these sub-centers. As shown in Figure 2, the distances between the anchor feature $\mathbf{x}_i$ and its corresponding cross-view sub-centers can be calculated as follows:

$$D(\mathbf{x}_i, \mathbf{c}_{y_ia_k}) = \|\mathbf{x}_i - \mathbf{c}_{y_ia_k}\|_2 \quad (9)$$

The largest distance among the angles is selected as the loss. Other angles with a smaller distance are masked. In this way, the loss function can be written as follows:

$$L_c = \frac{1}{2} \sum_{i=1}^m D(\mathbf{x}_i, \mathbf{c}_{y_ia_k})^2$$

$$s.t. \arg\max_{a_k, a_k \neq a_i} D(\mathbf{x}_i, \mathbf{c}_{y_ia_k})$$

The update strategy in Equation 7 can be seen as an efficient measure to learn the centers, especially for more centers or complex loss functions. However, it is not responsible for the performance. The hard mining strategy learns better intra-class compactness and thus could be beneficial to the performance. Comparisons among original center loss, simplified center loss and angle center loss will be given in Table II.

For gait recognition, there are often a limited number of training samples for each identity. This may cause severe

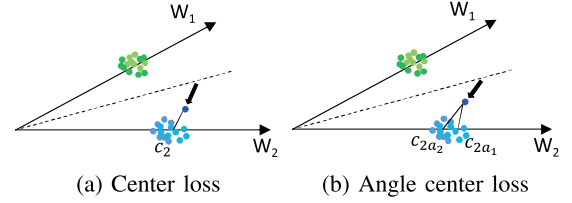


Fig. 2. Comparison between center loss and angle center loss.

overfitting for classification based methods. By modeling the metrics inside each identity, we are able to learn better intra-subject distributions compared with center loss. Since the sub-centers are correlated with the angles, the proposed loss function is called angle center loss (ACL). In [63] a novel loss function was also proposed for cross-view human re-identification by introducing sub-centers for each identity. The main difference is the update strategy of the sub-centers. In that work, all sub-centers are updated through an iteration-based optimization. Our method only updates on the hardest sub-center with the stochastic gradient descent (SGD) algorithm. Thus, it learns better intra-subject variations and is more efficient.

E. Local Gait Features

Human body partition has been used in previous works, including both model-based methods and appearance-based methods. Lee and Grimson [64] partitioned the silhouettes into 7 regions and used ellipses to fit each region. Rida *et al.* [19] partitioned the human body into 4 parts by group Lasso to select the most discriminative human body parts. Rokanujjaman *et al.* [20] used a 5-part partition and claimed the importance of the head, waist and lower leg regions. These works relied on non-uniform partitions, and thus we are inspired to use multiple CNNs for different non-uniform parts.

Figure 3 illustrates the proposed local feature extraction network. We train multiple separate CNNs for each local part. The method is similar to the part-based convolutional baseline (PCB) model for person re-identification in [31]. The difference is 2-folded.

- (1) In [31], local features are extracted from a shared global backbone network. For gait recognition, we do not share

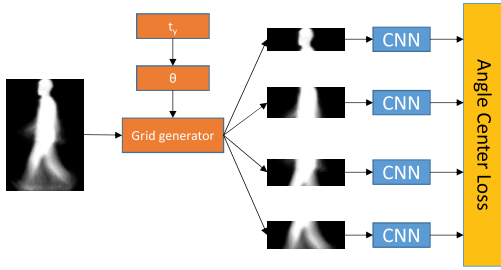


Fig. 3. Local feature extraction. The human body is partitioned into four different horizontal parts by simplified spatial transformer networks.

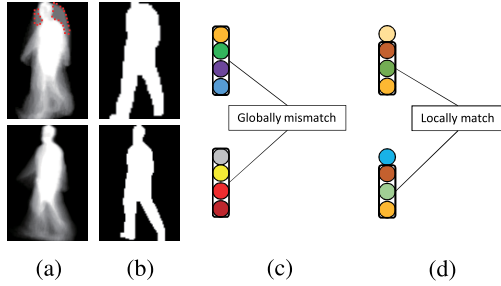


Fig. 4. Comparison between global features and concatenated local features. (a) Genuine GEI. Note the differences in the head region. (b) Genuine silhouettes. (c) Global features are mismatched. (d) Concatenated local features are locally matched.

the backbone network. The gait silhouette or GEI is partitioned into N horizontal parts at the image level. A classifier is then trained for each part. Note that these CNNs for different parts do not share parameters. The model has to classify the identity of the subject with only a local part rather than the whole body. The local features are concatenated as a final descriptor which can be seen as the hierarchical feature coding as described in [65].

- (2) We learn to localize the informative parts with a simplified spatial transformer network [56] rather than the uniform horizontal partition as proposed in [31].

There are two reasons for the equal combination of local features. (1) Avoid the problem of misalignment. In gait recognition, although the GEI/silhouettes are roughly aligned, there are still mismatching problems due to the low-quality segmentation. Pose estimation may be helpful with extra computation. As illustrated in Figure 4, global features may be completely different for a genuine pair while local features could preserve the similarities in the leg regions. (2) The local features are complementary to each other. The feature fusion could extract discriminative fine-grained features compared with global features. It should be noted that the equal combination is a trivial setting, which achieves promising results in person re-identification [31]. A further study on a weighted combination of different parts will be given in the discussion subsection.

The original spatial transformer network learns the affine transformation by a localization network. The transformation includes cropping, translation, rotation, scale, and skew. In [9], data augmentation approaches including rotation and scaling did not improve performance. The transformation is then

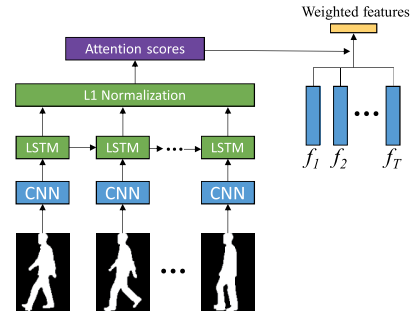


Fig. 5. Temporal attention model. Temporal attention score is predicted by the LSTM model for each frame. The scores are normalized along the time axis.

constrained as follows:

$$A_{\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & t_y \end{bmatrix} \quad (10)$$

where t_y represents vertical translation. The constrained transformation can be seen as a horizontal attention model which localizes informative parts of the human body. Since most gait silhouettes are normalized properly, there is no need to add additional localization networks to learn the transformation for each image as in [55]. We set only one parameter t_y for each horizontal localizer. With the differentiable image sampling algorithm [56], the translation parameter could be learned in an end-to-end manner. In this paper, 4 localizers have been selected. Using these spatial transformer networks directly from random initialization will cause divergence. Each localizer is initialized so that the attention part is the same as the hand-crafted uniform partition. Then the network could learn the deformable local attention parts. At testing time, the model extracts local features with the learned partition.

F. Temporal Gait Features

Although GEI is often used as the feature for gait recognition, it loses temporal information at the same time. In action recognition or video-based person re-identification, dynamic temporal features are important cues. Concatenated features along the temporal dimension can keep every detailed feature for each frame. The gait period of the gallery and probe cannot be aligned properly. Averaging features across the temporal dimension is a simple approach to get rid of the misalignment. The fused features can be regarded as another kind of GEI at the feature level. However, simply average pooling is not enough to capture the complex dynamic features since it only takes the average of several static features. We introduce an LSTM-based temporal attention model to learn the temporal score for each frame.

Figure 5 illustrates the proposed temporal attention model. We introduce long short-term memory (LSTM) units to learn the temporal attention since these units can memorize previous information of gait sequences. The dimension of the LSTM model is 1, which plays the role of the attention score. Since there are no direct labels for the attention mechanism, the temporal attention has to be learned from the data itself. Similar to [66], we extract frame-level features f_t by a 2DCNN, where f_t is the 2DCNN feature for frame t . Then a new branch of an

LSTM layer is added for attention. For an input gait sequence of T frames, the LSTM layer outputs activations of T frames as scores. These scores are then L1-normalized along the time axis as below.

$$a_t = \frac{s_t}{\sum_{i=1}^T s_i} \quad (11)$$

where a_t is the L1-normalized attention score for frame t , and s_i is the score value for frame i . The L1-normalization makes sure the sum of the attention scores for the sequence equals to 1.

With these temporal attention scores captured, the overall feature presentation for the gait sequence can be rewritten as follows.

$$f_s = \sum_{t=1}^T a_t f_t \quad (12)$$

where f_s represents the feature for the whole sequence. Note that when $a_t = 1/T$, the formulation degrades to average temporal pooling which treats each frame equally. With the help of the attention model, the algorithm pays more attention on those discriminative frames and thus improves the overall performance.

G. Training and Testing

The following part describes how to train and test with the proposed method.

1) *Training*: The input of the network is a sequence of silhouettes. A consecutive T frames of silhouettes are regarded as a sample in our setting. We randomly select samples from the whole gait sequences. This can be seen as a temporal data augmentation approach. These samples go through a 2DCNN to extract frame-level features. The features are fused temporally through an LSTM-based attention model. Then loss functions can be added to supervise learning.

2) *Testing*: The gait sequences are of varied length at testing time. Since the LSTM-based temporal attention model does not rely on a fixed input length, one feature vector can be generated for a gait sequence regardless of the sequence length. With the spatial-temporal features extracted for each gait sequence, the matching metrics can be calculated using the cosine similarities between these feature vectors.

IV. EXPERIMENTS

In this section, we first evaluate the impact of each aspect of gait recognition to find out the best settings. Then the proposed method is evaluated on several cross-view gait recognition datasets: CASIA-B, OULP and OUMVLP. These three datasets provide us with silhouettes, which could be helpful for better temporal modeling. To better understand the effect of different loss functions qualitatively, we visualize the learned gait features.

A. Datasets

1) CASIA-B [67] is a small dataset with 124 subjects. Each subject has 110 sequences, namely 11 different views ($0^\circ, 18^\circ, \dots, 180^\circ$) and 10 sequences per subject for each view.

Among the 10 sequences, 6 are taken under normal conditions (NM). The first 4 sequences (NM01-NM04) form gallery sets and the rest 2 sequences (NM05-NM06) are used as probes. Apart from the normal condition sequences, 2 sequences are taken when the subjects are in their coats (CL01-CL02) and 2 sequences are taken with bags (BG01-BG02). The dataset enables researchers to study the cross-view and cross-clothing problems at the same time.

2) OULP [68] contains 4,007 subjects (2,135 males and 1,872 females) with only 4 view angles ($55^\circ, 65^\circ, 75^\circ, 85^\circ$). Two sequences are available for each subject, one for the gallery and the other for the probe. Compared with CASIA-B, OULP has smaller view differences and no variants in walking conditions. However, the large number of subjects enables us to compare different gait recognition approaches with statistical significance.

3) OUMVLP [69] is a recently published large-scale large view-variation gait database. It includes more than 10,000 subjects with 14 view variations ($0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ, 180^\circ, 195^\circ, 210^\circ, 215^\circ, 240^\circ, 255^\circ$, and 270°). Similar to OULP, only two sequences are provided, one for the gallery and the other for the probe. Following [69], only four typical view angles ($0^\circ, 30^\circ, 60^\circ$, and 90°) are evaluated. The dataset can provide us with stable comparison results.

B. General Settings

To find out the best settings of loss functions and spatial-temporal feature learning, we experiment on CASIA-B and OUMVLP cross-view gait recognition. For CASIA-B, we keep the same train/test setting as that of [9]. The first 74 subjects are used for training and the left 50 people for testing. For training, the first 6 sequences (NM01-NM06) of one subject are used. At testing time, the first 4 sequences (NM01-NM04) are used as gallery and the other two sequences (NM05-NM06) as probes. For OUMVLP, the recognition accuracy is evaluated in two tasks: identification and verification. Rank-1 identification accuracy is used to measure the identification performance. Equal error rates (EERs) are used as a criterion of the verification capability.

The models are trained with 8 Nvidia Titan V GPUs, and the optimizer is Stochastic Gradient Descent (SGD). We list the settings for the learning rate and the batch size in Table I. Generally speaking, a learning rate of 0.1 accelerates the training process with the help of batch normalization. For the case without batch normalization, the initial learning rate is set to 0.01. Since we use data parallelism for computing, these values should be multiplied by the number of GPUs. That is, the learning rate for GEI models with BN is $0.1 \times 8 = 0.8$ in our case.

C. Backbone Architecture

Deeper networks have shown great improvements on image classification, object detection, face recognition and person re-identification. However, current gait recognition methods usually switch to shallow networks [9]–[11]. The reason behind this phenomenon might be the fact that gait has less texture information compared with other biometrics. In this

TABLE I
IMPLEMENTATION DETAILS

Experimental settings	Batch size	Learning rate
GEI(with BN)	128	0.1
GEI(without BN)	128	0.01
silhouette(with BN)	32	0.1
silhouette(without BN)	32	0.01



Fig. 6. Network backbone. It is similar to the architecture in [9]. The main difference is the use of batch normalization, which proves to be important in our experiments.

paper, we follow the network architecture of [9] with some modifications. The local response normalization (LRN) layers are removed and a pooling layer *pool3* is added to further down-sample the feature dimension. The convolutional layers are padded with a size of 3×3 . By doing so, the output shape of *pool3* is $256 \times 16 \times 11$. Motivated by the success of the additional block in [28], we also add a block of 4 additional layers as “BN-Dropout-BN-FC” as shown in Figure 6. The batch normalization layers can be merged at the inference stage, which makes sure the computation cost is similar to the models in [9] and [11]. For CASIA-B the feature dimension is 128 and for OULP and OUMVLP the feature dimension is 512. For OUMVLP, we modify the backbone network to fit the large dataset. The last pooling layer *pool3* is removed to capture more spatial features. With the similar backbone architecture, the additional performance boost would come from the proposed loss function, the learned local partition and the temporal attention model.

D. Impact of Loss Functions

To validate the effectiveness of the proposed loss function, we fix the input as the GEI of the whole body. Several state-of-the-art loss functions are evaluated including softmax* loss (softmax without batch normalization), softmax loss, COCO loss, CosFace loss, ArcFace loss, center loss and the proposed ACL loss. The hyper-parameters are selected by grid search. COCO loss, CosFace loss and ArcFace loss share the scale factor of 20. The angular margin for CosFace and ArcFace are selected according to the original paper, that is, 0.35 for CosFace and 0.5 for ArcFace. For center loss and ACL loss, the factor λ is chosen as 0.003 following [25].

Table II lists the evaluation results for CASIA-B cross-view identification and OUMVLP identification/verification. We can draw the following conclusions:

- Batch normalization plays an important role for bare softmax loss. The performance improves a lot since batch normalization regularizes the data distribution.

TABLE II
RANK-1 ACCURACIES (%) AND EQUAL ERROR RATES (EERs) (%) WITH DIFFERENT LOSS FUNCTIONS. FOR CASIA-B, WE REPORT CROSS-VIEW AVERAGE ACCURACIES EXCLUDING IDENTICAL-VIEW CASES. FOR OUMVLP, IDENTICAL-VIEW CASES ARE INCLUDED IN THIS TABLE

Loss function	CASIA-B	OUMVLP	
	Acc	Acc	EERs
softmax*	67.0	46.9	5.05
softmax	82.4	72.3	1.97
COCO	83.0	72.4	1.71
CosFace	83.1	72.9	2.10
ArcFace	83.9	73.2	1.85
center	86.0	74.0	1.63
simplified center	86.0	74.0	1.65
ACL	86.3	74.3	1.63

Furthermore, the use of batch normalization enables us with a larger learning rate.

- Feature normalization based methods including COCO loss, CosFace loss and ArcFace loss are sensitive to the hyper-parameters. Tuning a suitable scale factor needs a careful grid search.
- Center loss is robust with the recommended parameter and achieves promising performance.
- The simplified center loss with the update strategy in Equation 7 achieves similar performance with the original center loss.
- The proposed ACL loss outperforms the above loss functions in terms of identification and verification. Since it shares the same update strategy with the simplified center loss, the performance gain comes from the hard mining strategy.

It should be noted that CASIA-B contains only 50 subjects for evaluation while OUMVLP contains 5,154. The proposed loss function has not achieved much improvement on the large-scale dataset OUMVLP. The reason is two folded. (1) The information loss in GEIs. Without effective spatial-temporal feature learning, we cannot expect promising results. (2) The use of the extra block in the backbone sets a strong baseline for gait recognition. This may shorten the performance distance between center loss and ACL loss.

The hard positive strategy is effective when there are unbalanced samples in some angles. To validate this, we only keep a subset of GEI samples per subject from 90° in CASIA-B at training time. The evaluation is the same to the normal cases without data removal. Figure 7 illustrates the comparison between different loss functions with an unbalanced number of views. It can be seen that center loss and ACL loss still performs well with unbalanced data. The proposed ACL loss further boosts the performance compared with center loss. The method is beneficial since some views of gait data may be lost at the capturing stage.

E. Impact of Local Features

To study the impact of local features, we fix the input as GEI and the loss function as softmax loss. We set the global features with a dimension of $N = 128$ as the baseline.

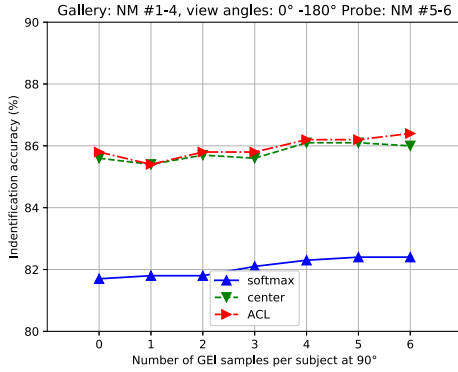


Fig. 7. Rank-1 accuracies (%) when removing some of GEI samples per subject from 90° at training time.

TABLE III
CASIA-B CROSS-VIEW AVERAGE ACCURACIES (%) WITH
DIFFERENT PARTITION SETTINGS

Partition methods	Accuracy
global features	82.4
2 uniform partition	86.2
3 uniform partition	87.0
4 uniform partition	87.4
5 uniform partition	85.2
4 learned partition	88.0
4 spatial attention	87.7

Several uniform partition settings are tested. For the model with $P(P = 1, 2, 3, 4, 5)$ parts, the local feature dimension is set as N/P to make sure the final feature dimension is roughly the same. At testing time, the feature vectors of these parts are extracted separately and then concatenated into a single descriptor.

Table III lists the results for different parts. Concatenated local features from horizontal parts achieve better results compared with global features. Among all the uniform partition settings, the 4-part partition achieves the best performance. Since the images are partitioned at the image level, the partition is interpretable as head, upper limb, thigh, and lower leg.

Although simple and effective, the above partition is indeed a hand-crafted uniform partition. A better approach would be a learned localization of informative parts. The proposed simplified spatial transformer network learns the discriminative local parts during training. At testing time, the partition is fixed without additional computation. From Table III, it can be seen that the simplified spatial transformer network achieves better performance compared with the uniform partition. We find that the network tends to move its attention from lower legs to the upper legs, which means that upper legs are likely more important than lower legs.

Spatial attention mechanism is also used to localize the informative human parts with irregular shape in a “soft” manner. It should be noted that the silhouettes/GEI can be regarded as a kind of attention. Body parts could be hardly localized in person re-identification while easily focused in gait recognition. A comparison with spatial attention mechanism is listed in Table III. Additional networks are proposed

TABLE IV
CASIA-B CROSS-VIEW AVERAGE ACCURACIES (%) WITH THE
COMBINATION OF DIFFERENT LOCAL PARTS

Part combination	Accuracy
part 1	59.0
part 2	48.8
part 3	52.6
part 4	46.3
part 1+2	75.6
part 3+4	67.6
part 1+3	78.1
part 2+4	69.8
part 1+2+3+4	88.0

to learn the spatial attention score at the image level. There is no significant improvement over the learned “hard” partition.

We further study which part is more important among the learned 4 parts. Table IV lists the accuracy for a single part and their combinations. We can conclude that head and upper leg regions are important while the lower legs contribute the lowest accuracy for cross-view gait recognition. By fixing the gallery and probe views as 90°, the lower legs achieves an accuracy of 99% which is better than 97% of the upper legs. This indicates that lower legs are more important for non-cross-view gait recognition. The conclusion is similar to that of [20] which also emphasized the importance of head and leg regions. However, their work studies cross-clothing gait recognition which has no view differences. The proposed method focuses on cross-view gait recognition and thus the important local regions would be view invariant. The lower parts of legs with unstable features are neglected by the networks.

F. Impact of Temporal Features

To study the impact of temporal features, we use the global silhouettes with softmax loss. The window size is chosen as long as 24 frames. The reason is two-folded. (1) It is roughly the period of a gait sequence. All possible body poses are included in such a long temporal window size. It is enough for the LSTM-based attention model to learn the dynamic features. (2) A window size of 24 can be seen as the GEI at the feature level. The hyper-parameters from GEI-based experiments could be transferred to the multi-frame silhouette cases.

Table V shows the performance of different temporal fusion methods. Although simple average pooling achieves promising results compared with GEI, the LSTM-based attention model could further boost performance.

A stack of LSTM layers can be used to learn better temporal features for action recognition. However, gait recognition computes similarities based on the features while action recognition is mainly a classification problem. The LSTM units here only output one-dimension feature as the attention score for each frame rather than high-dimension dynamic features. We list the comparison of different number of LSTM layers in Table V. As can be seen, more LSTM layers do not improve results.

Figure 8 illustrates the attention scores predicted by the model. Low attention scores are given to those frames with low

TABLE V
CASIA-B CROSS-VIEW AVERAGE ACCURACIES (%) WITH
DIFFERENT TEMPORAL FUSION METHODS

Fusion methods	Accuracy
GEI	82.4
Average pooling	88.7
Temporal attention	89.1
2-layer LSTM	89.1
3-layer LSTM	89.1

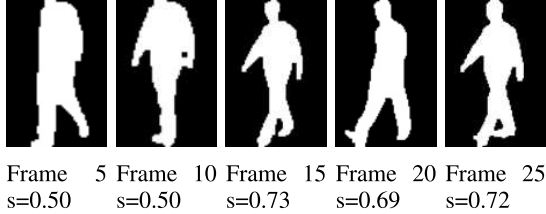


Fig. 8. Visualization for temporal attention. Silhouettes of bad quality are less focused by the temporal attention model.

quality, e.g., silhouettes without head regions because of simple background subtraction preprocessing algorithms. These silhouettes may not contribute enough to the identification accuracy and should be neglected. The traditional temporal average pooling method could not avoid these low-quality silhouettes since every frame is treated equally.

G. Feature Distribution

At testing time, we compute the cosine similarities between the feature vectors. The feature distribution could be studied for a better understanding. In this section, we study the effect of different loss functions qualitatively.

A toy example with a feature dimension of 3 is constructed. 8 subjects (ID001-ID008) in CASIA-B are selected. Since there are a limited number of GEI samples for each identity, we use raw silhouettes from all angles as input. Furthermore, we do not test on other subjects since such a low feature dimension cannot guarantee the generalization for other subjects. The visualization can be achieved by directly taking the output feature vectors.

Figure 9 lists the results of the toy example. Several models have been trained with different loss functions. Traditional softmax loss without batch normalization (softmax*) generates the worst distribution. The decision boundaries are ambiguous. Softmax loss with batch normalization works better compared with softmax*. However, the features are not discriminative enough since softmax loss only learns separable features. COCO loss, CosFace loss and center loss get similar distributions. ArcFace loss fails to convergence with such a low feature dimension. The proposed loss function generates better distributions than other loss functions.

H. Experiments on CASIA-B

Experiments have been conducted on CASIA-B cross-view gait recognition following the same data split setting as that of [9]. Table VI illustrates the results. By combining

these three aspects, we achieve the state-of-the-art results for cross-view gait recognition, outperforming Wu *et al.* [9] which relied on an ensemble of networks and Chao *et al.* [45] with a very high feature dimension.

I. Experimental Results on OULP

Experiments are then conducted on OULP with over 4,000 subjects following the five-fold experimental setting of [9]. We use one set for testing and keep the others for training for each run. In detail, 3,075 unique subjects are used for training and 768 subjects are used for testing.

Table VII lists the detailed results for each view. The proposed method achieves the highest accuracy for both cross-view and identical view cases. Results have also been listed in Table VIII following the evaluation metrics in [11]. We can get the following conclusions:

- The baseline loss function softmax* achieves better performance than GEINet, which also relies on softmax loss without batch normalization. In [10], the authors used the L2 distance between features as the similarity metric. It has been proved in [26] that cosine similarity is more suitable for softmax loss.
- Spatial-temporal features introduce limited improvement. This is reasonable since OULP has relatively small view differences.

J. Experimental Results on OUMVLP

To prove the effectiveness of the proposed method, we also experiment on OUMVLP. We follow the same evaluation setting as that of [11], that is, 5,153 people for training and 5,154 people for testing. Silhouettes from four typical views are evaluated for cross-view recognition. The dimension for both global and local features is selected as 512 and thus the concatenated feature dimension is 2,048.

As listed in Table IX, our method outperforms other methods greatly with only global GEI as input following the evaluation metrics in [11]. This shows the effectiveness of the proposed loss function. We also list the results at stricter criteria by excluding identical-view cases in Table X and Table XI. It can be seen that the proposed method performs well for silhouettes at the same time, which shows the effectiveness of the spatial-temporal feature learning method.

K. Ablation Study

We conduct ablation study on the three parts of the whole framework, i.e., ACL, LSTM attention model and body-part localization to evaluate how each part boosts the performances in Table XII. We can get the following conclusions:

- The proposed loss function is robust in the cases of using different spatial-temporal features. It can always improve performance over center loss.
- The combination of local features achieves promising results, especially about 6% improvement when using the temporal attention model.
- The LSTM attention model further boosts performance. The improvement becomes larger with the local features.

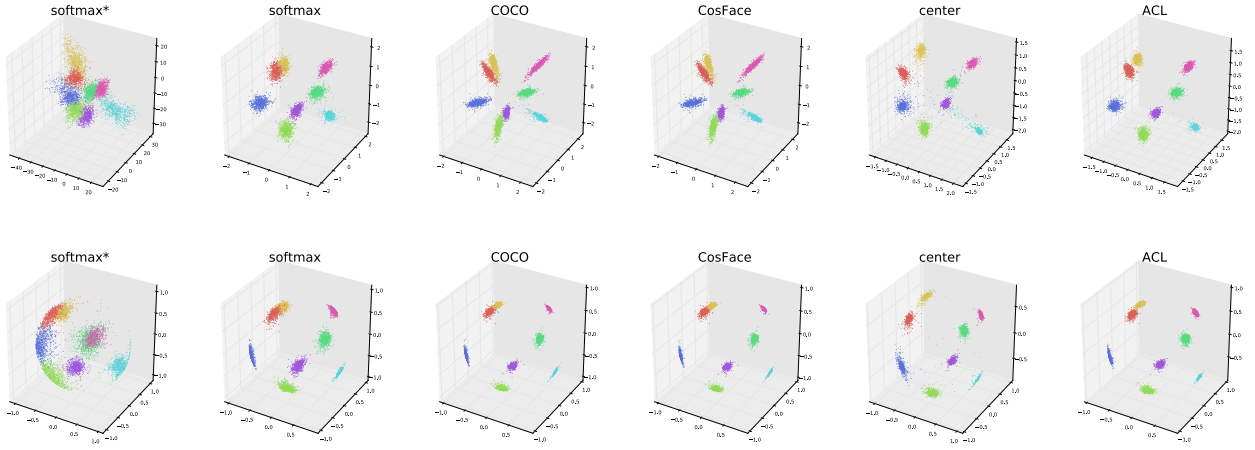


Fig. 9. Feature distribution of several state-of-the-art loss functions on a toy example with a feature dimension of 3. The model is trained with the first ID001-ID008 in the CASIA-B dataset. The first row maps the 3D features onto the Euclidean space, while the second row projects the 3D features onto the angular space. The inter-class and intra-class variations can be observed.

TABLE VI

CASIA-B CROSS-VIEW AVERAGE ACCURACIES (%) TRAINED BY THE FIRST 74 SUBJECTS. IDENTICAL-VIEW CASES ARE NOT INCLUDED

Gallery NM #1-4	0°-180°										
Probe NM #5-6	0°	18°	36°	54°	72°	90°	108°	126°	144°	162°	180°
ViDP [40]	-	-	-	64.2	-	60.4	-	65.0	-	-	-
PEI+MGANs [18]	-	-	-	84.2	-	72.3	-	83.0	-	-	-
3D MT network [9]	87.1	93.2	97.0	94.6	90.2	88.3	91.1	93.8	96.5	96.0	85.7
Ensemble of networks [9]	88.7	95.1	98.2	96.4	94.1	91.5	93.9	97.5	98.4	95.8	85.6
GaitSet [45]	90.8	97.9	99.4	96.9	93.6	91.7	95.0	97.8	98.9	96.8	85.8
ACL+local+temporal	92.0	98.5	100.0	98.9	95.7	91.5	94.5	97.7	98.4	96.7	91.9

TABLE VII

OULP CROSS-VIEW AVERAGE ACCURACIES (%) FOR ALL PAIRS OF FOUR VIEW ANGLES

Probe angle	Method	Gallery angle					Identical angle
		55°	65°	75°	85°	mean	
55°	Wu et al. [9]	-	98.3±0.1	96.0±0.1	80.5±0.4	91.6±0.2	98.8±0.1
	He et al. [18]	-	99.4±0.1	96.1±0.3	77.9±0.4	91.6±0.2	98.8±0.1
	Iwama et al. [68]	-	-	-	-	-	84.7
	proposed	-	99.6±0.4	99.6±0.2	99.0±0.2	99.4±0.4	99.3±0.4
65°	Wu et al. [9]	96.3±0.2	-	97.3±0.0	83.3±0.3	92.3±0.1	98.9±0.2
	He et al. [18]	97.7±0.1	-	98.5±0.1	84.4±0.5	-	-
	Iwama et al. [68]	-	-	-	-	-	86.6
	proposed	99.3±0.5	-	99.8±0.1	99.7±0.2	99.6±0.4	99.6±0.4
75°	Wu et al. [9]	94.2±0.2	97.8±0.2	-	85.1±0.2	92.4±0.1	98.9±0.0
	He et al. [18]	94.8±0.1	98.9±0.1	-	86.4±0.3	-	-
	Iwama et al. [68]	-	-	-	-	-	86.9
	proposed	98.8±0.4	99.4±0.3	-	99.6±0.2	99.3±0.5	99.6±0.1
85°	Wu et al. [9]	90.0±0.5	96.0±0.3	98.4±0.1	-	94.8±0.3	98.9±0.1
	He et al. [18]	86.9±0.6	97.4±0.3	99.5±0.1	-	-	-
	Iwama et al. [68]	-	-	-	-	-	85.7
	proposed	98.0±0.5	99.0±0.3	98.5±0.2	-	98.8±0.7	99.7±0.1

L. Efficiency

As discussed in [70], the efficiency of the pair-wise similarity learning method [9] is limited. On the other hand, the feature learning methods such as GaitSet [45] only need to extract features once and then compute the similarities between those features. Table XIII lists the efficiency comparison on CASIA-B and OUMVLP.

The model with LSTM attention is weak in efficiency because of the low parallel computing capacity with varied testing sequence length. However, for real-time cases, the LSTM model does not cost much extra computation. The

feature dimension of our model is 2,048 for OUMVLP, which is much smaller than [45] with a dimension of 15,872. The low dimension makes it possible for fast retrieval. We also list the cases without LSTM attention. The temporal features are fused by average pooling. This setting enables us to obtain better efficiency.

M. Discussion

1) *Loss Functions*: Deep learning has the advantage of learning discriminative features or metrics. Previous gait recognition methods often rely on classic loss functions,

TABLE VIII

CROSS-VIEW GAIT RECOGNITION RESULTS OBTAINED ON OULP WITH OUR METHOD AND COMPARISON WITH OTHER METHODS REPORTED BY [11]. SOFTMAX* MEANS SOFTMAX LOSS WITHOUT BATCH NORMALIZATION

(a) Rank-1 identification rates (%)						(b) EERs (%)					
method	Angular Difference				mean	method	Angular Difference				mean
	0°	10°	20°	30°			0°	10°	20°	30°	
DM	91.5	49.5	11.2	2.8	44.6	DM	4.3	8.4	20.2	31.3	13.2
LDA	97.8	97.1	93.4	82.9	94.6	LDA	2.1	2.5	3.7	5.7	3.1
VTM	91.5	64.0	37.2	20.5	58.7	VTM	4.3	10.5	14.8	18.9	11.1
GEINet	96.5	95.8	92.5	84.9	93.8	GEINet	1.9	2.1	3.0	4.9	2.6
Wu[9]	98.9	95.5	92.4	85.3	94.3	Wu[9]	-	-	-	-	-
3in+2diff	99.2	99.2	98.6	97.0	98.8	2in+diff	0.2	0.2	0.2	0.4	0.2
softmax*+global+GEI	96.9	96.0	92.1	82.3	93.5	softmax*+global+GEI	0.95	0.96	1.20	2.01	1.15
softmax+global+GEI	99.2	99.2	98.7	97.3	98.8	softmax+global+GEI	0.35	0.36	0.42	0.58	0.40
center+global+GEI	99.2	99.2	99.0	97.7	99.0	center+global+GEI	0.35	0.32	0.36	0.52	0.36
ACL+global+GEI	99.4	99.4	99.2	98.3	99.2	ACL+global+GEI	0.29	0.27	0.30	0.38	0.30
ACL+local+temporal	99.5	99.5	99.3	98.5	99.3	ACL+local+temporal	0.19	0.21	0.25	0.30	0.23

TABLE IX

CROSS-VIEW GAIT RECOGNITION RESULTS OBTAINED ON OUMVLP WITH OUR METHOD AND THE COMPARISON WITH OTHER METHODS REPORTED BY [11]. SOFTMAX* MEANS SOFTMAX LOSS WITHOUT BATCH NORMALIZATION

(a) Rank-1 identification rates (%)						(b) EERs (%)					
method	Angular Difference				mean	method	Angular Difference				mean
	0°	30°	60°	90°			0°	30°	60°	90°	
DM	77.4	2.4	0.2	0.0	20.3	DM	6.5	25.2	41.4	46.2	27.2
LDA	81.6	10.1	0.8	0.1	24.4	LDA	6.2	22.7	35.7	40.1	24.0
VTM	77.4	2.7	0.6	0.2	20.5	VTM	6.5	26.8	34.2	38.5	25.0
GEINet	85.7	40.3	13.8	5.4	40.7	GEINet	2.4	5.9	12.7	17.2	8.1
LB(Wu)	89.9	42.2	15.2	4.5	42.6	LB(Wu)	1.0	3.3	6.7	9.3	4.3
MT(Wu)	89.3	49.0	20.9	8.2	46.9	MT(Wu)	0.9	2.5	5.2	7.0	3.3
3in+2diff	89.5	55.0	30.0	17.3	52.7	2in+diff	1.0	2.0	3.4	4.2	2.4
softmax*+global+GEI	91.6	49.9	17.5	7.0	46.9	softmax*+global+GEI	1.39	3.74	7.99	10.45	5.05
softmax+global+GEI	95.7	81.7	54.0	33.7	72.3	softmax+global+GEI	0.67	1.50	2.93	4.06	1.97
center+global+GEI	95.5	82.4	57.6	39.0	74.0	center+global+GEI	0.64	1.27	2.40	3.20	1.63
ACL+global+GEI	95.8	82.9	57.4	38.9	74.3	ACL+global+GEI	0.63	1.24	2.39	3.29	1.63
ACL+local+temporal	99.5	95.8	77.1	66.3	88.3	ACL+local+temporal	0.18	0.57	1.52	1.85	0.87

TABLE X

OUMVLP RESULTS EXCLUDING IDENTICAL-VIEW CASES UNDER FOUR TYPICAL VIEWS

method	Probe				Mean
	0°	30°	60°	90°	
Gaitset(silhouettes)	77.7	86.9	85.3	83.5	83.4
proposed(silhouettes)	71.6	95.0	85.1	86.7	84.6
DiGGAN(GEI)	48.9	62.3	59.1	57.8	57.0
proposed(GEI)	49.0	79.3	71.6	68.3	67.1

e.g., softmax loss, contrastive loss and triplet loss. Although gait recognition and face recognition have similarities, we believe there should be gait-related loss functions for the specific setting, e.g., cross-view and cross-walking-condition settings. These gait-related loss functions could borrow ideas from other fields such as face recognition and person re-identification.

Softmax loss has been widely used in face recognition, person re-identification and action recognition. It could be regarded as a strong baseline for gait recognition, especially with the extra block in the backbone. We could also compare with other methods based on softmax loss, such as GEINet.

The proposed angle center loss originates from center loss, which performs well for gait recognition. The introduction of

TABLE XI

OUMVLP RESULTS EXCLUDING IDENTICAL-VIEW CASES UNDER ALL VIEWS

Probe	Galley All 14 Views		
	GEINet	GaitSet	proposed
0°	11.4	79.5	74.0
15°	29.1	87.9	88.3
30°	41.5	89.9	94.6
45°	45.5	90.2	95.4
60°	39.5	88.1	88.0
75°	41.8	88.7	91.3
90°	38.9	87.8	90.0
180°	14.9	81.7	76.7
195°	33.1	86.7	89.5
210°	43.2	89.0	95.0
225°	45.6	89.3	94.9
240°	39.4	87.2	88.0
255°	40.5	87.8	90.8
270°	36.3	86.2	89.8
mean	35.8	87.1	89.0

sub-centers within a class makes it possible to push one anchor feature to its corresponding cross-view sub-center rather than to the average class center. The operation follows the notion of cross-view gait recognition and thus could learn better intra-subject compactness. The proposed method is useful when the gait views are unbalanced at training time. This is

TABLE XII

ABLATION STUDY ON CASIA-B TRAINED BY THE FIRST 74 SUBJECTS. IDENTICAL-VIEW CASES ARE NOT INCLUDED

Loss type		Spatial features		Temporal features		Mean
center	ACL	global	local	GEI	LSTM attention	
✓		✓		✓		86.0
✓			✓	✓		88.5
	✓	✓		✓		86.3
	✓		✓	✓		88.7
✓		✓			✓	89.3
✓			✓		✓	95.8
	✓	✓			✓	90.0
	✓		✓		✓	96.0

TABLE XIII

EVALUATION EFFICIENCY COMPARISON ON CASIA-B AND OUMVLP. THE RESULTS FOR THE UPPER TWO METHODS ARE REPORTED BY [70] WITH 4 TITAN GPUS. 8 TITAN GPUS ARE USED TO EVALUATE OUMVLP

Method	CASIA-B	OUMVLP(All views)
Wu et al.[9]	40 min	-
Zhang et al.[70]	3 min	-
Chao et al.[45]	-	7 min
proposed	1.5 min	45 min
proposed(no lstm)	0.15 min	7.7 min

beneficial since some views of gait data may be lost at the capturing stage.

2) *Data Cleaning*: In our experiments on CASIA-B, current state-of-the-art feature normalization based methods perform similarly to softmax loss. The same phenomenon can be observed in LFW, a small-scale face recognition dataset where traditional softmax loss also achieves promising results [26]. The reason behind this phenomenon could be noise in the training data. For face recognition, data cleaning could further boost performance as opposed in [26] and [28]. However, silhouettes generated from background subtraction algorithms are often in bad quality. There could be two solutions. One is to manually modify the silhouettes as opposed in [68]. This could guarantee the quality of the silhouettes at the cost of much time. On the other hand, deep learning based pedestrian detection and tracking, human segmentation and human pose estimation algorithms could be used to generate high-quality silhouettes automatically. We think deep learning based approaches could be an efficient method for gait data cleaning.

3) *Part Combination*: Equally concatenated horizontal features are commonly used in previous works. The importance of these parts could be different for gait recognition. To achieve this, we learn an importance score ranging from 0-1 for each part. These scores are fixed at testing time to output a weighted combination of different parts. As listed in Table XIV, the weighted combination of different parts further improves performance. After normalizing the scores, we find that head region counts most while lower leg region counts least.

We also experiment with the hierarchical combination of the local features. The local features from different partitions ($P=1,2,3,3,5$) are concatenated into a final descriptor with a feature dimension of roughly $128 \times 5 = 640$. The hierarchical

TABLE XIV

CASIA-B CROSS-VIEW AVERAGE ACCURACIES (%) WITH COMPLEX PART COMBINATION STRATEGIES

Part combination	Accuracy
equal combination	88.0
weighted combination	88.4
hierarchical combination	93.1

combination outperforms all other simple combinations by a large margin. However, the computation cost and feature dimension become higher.

V. CONCLUSION

This paper has studied discriminative spatial-temporal feature learning for cross-view gait recognition. A robust, effective and gait-related loss function is proposed with the following advantages.

- Compared with feature normalization based methods which have the problem of tuning hyper-parameters, the proposed loss function is robust to different local parts and window sizes. Compared with contrastive loss or triplet loss, the proposed loss function avoids hard negative mining problems.
- The proposed loss function achieves better performance compared with other state-of-the-art loss functions. Feature distribution visualization proves the discrimination qualitatively.
- Sub-centers have been introduced for each angle of the same identity. Hard positive pairs can be mined by calculating the distances between the anchor feature and the corresponding cross-view sub-centers.

Further improvements have been achieved by cooperating spatial-temporal information. A simplified spatial transformer network is proposed to automatically localize informative parts of the human body and has achieved better performance than the uniform partition. An LSTM-based temporal attention model is also proposed to better learn the temporal gait features compared with temporal mean average pooling.

The proposed loss function follows the notion of cross-view gait recognition and thus achieves better intra-subject compactness. It is also beneficial with unbalanced views of data at training time. The proposed method could capture detailed spatial-temporal gait features at the same time. By combining the above aspects, we achieve the state-of-the-art performance on several benchmarks including CASIA-B, OULP and OUMVLP. Thus, the proposed method could be a further step for cross-view gait recognition.

REFERENCES

- [1] P. K. Larsen, E. B. Simonsen, and N. Lynnerup, "Gait analysis in forensic medicine," *J. Forensic Sci.*, vol. 53, no. 5, pp. 1149–1153, 2008.
- [2] I. Bouchrika, M. Goffredo, J. Carter, and M. Nixon, "On using gait in forensic biometrics," *J. Forensic Sci.*, vol. 56, no. 4, pp. 882–889, 2011.
- [3] J. Man and B. Bhanu, "Individual recognition using gait energy image," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 2, pp. 316–322, Feb. 2006.
- [4] G. Zhao, G. Liu, H. Li, and M. Pietikainen, "3D gait recognition using multiple cameras," in *Proc. Int. Conf. Autom. Face Gesture Recognit.*, 2006, pp. 529–534.

- [5] G. Ariyanto and M. S. Nixon, "Model-based 3D gait biometrics," in *Proc. Int. Joint Conf. Biometrics*, 2011, pp. 1–7.
- [6] D. L. Swets and J. Weng, "Using discriminant eigenfeatures for image retrieval," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 8, pp. 831–836, Aug. 1996.
- [7] R. Martín-Félez and T. Xiang, "Uncooperative gait recognition by learning to rank," *Pattern Recognit.*, vol. 47, no. 12, pp. 3793–3806, Dec. 2014.
- [8] Z. Wu, Y. Huang, and L. Wang, "Learning representative deep features for image set analysis," *IEEE Trans. Multimedia*, vol. 17, no. 11, pp. 1960–1968, Nov. 2015.
- [9] Z. Wu, Y. Huang, L. Wang, X. Wang, and T. Tan, "A comprehensive study on cross-view gait based human identification with deep CNNs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 2, pp. 209–226, Feb. 2016.
- [10] K. Shiraga, Y. Makihara, D. Muramatsu, T. Echigo, and Y. Yagi, "GEINet: View-invariant gait recognition using a convolutional neural network," in *Proc. IEEE Int. Conf. Biometrics*, Jun. 2016, pp. 1–8.
- [11] N. Takemura, Y. Makihara, D. Muramatsu, T. Echigo, and Y. Yagi, "On input/output architectures for convolutional neural network-based cross-view gait recognition," *IEEE Trans. Circuits Syst. Video Technol.*, to be published.
- [12] C. Wang, J. Zhang, L. Wang, J. Pu, and X. Yuan, "Human identification using temporal information preserving gait template," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2164–2176, Nov. 2012.
- [13] X. Ben, C. Gong, P. Zhang, X. Jia, Q. Wu, and W. Meng, "Coupled patch alignment for matching cross-view gaits," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3142–3157, Jun. 2019.
- [14] X. Ben, C. Gong, P. Zhang, R. Yan, Q. Wu, and W. Meng, "Coupled bilinear discriminant projection for cross-view gait recognition," *IEEE Trans. Circuits Syst. Video Technol.*, to be published.
- [15] S. Yu, H. Chen, Q. Wang, L. Shen, and Y. Huang, "Invariant feature extraction for gait recognition using only one uniform model," *Neuro-computing*, vol. 239, pp. 81–93, May 2017.
- [16] I. J. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
- [17] S. Yu, H. Chen, E. B. G. Reyes, and N. Poh, "GaitGAN: Invariant gait feature extraction using generative adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 532–539.
- [18] Y. He, J. Zhang, H. Shan, and L. Wang, "Multi-task GANs for view-specific feature learning in gait recognition," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 1, pp. 102–113, Jan. 2019.
- [19] I. Rida, X. Jiang, and G. L. Marcialis, "Human body part selection by group lasso of motion for model-free gait recognition," *IEEE Signal Process. Lett.*, vol. 23, no. 1, pp. 154–158, Jan. 2016.
- [20] M. Rokanujjaman, M. A. Hossain, and M. R. Islam, "Effective part selection for part-based gait identification," in *Proc. Int. Conf. Elect. Comput. Eng.*, 2012, pp. 17–19.
- [21] T. Wolf, M. Babae, and G. Rigoll, "Multi-view gait recognition using 3D convolutional neural networks," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2016, pp. 4165–4169.
- [22] D. Thapar, A. Nigam, D. Aggarwal, and P. Agarwal, "VGR-Net: A view invariant gait recognition network," in *Proc. IEEE Int. Conf. Identity, Secur., Behav. Anal. (ISBA)*, Jan. 2018, pp. 1–8.
- [23] Y. Feng, Y. Li, and J. Luo, "Learning effective gait features using LSTM," in *Proc. IEEE Int. Conf. Pattern Recognit.*, Dec. 2016, pp. 325–330.
- [24] R. Liao, C. Cao, E. B. Garcia, S. Yu, and Y. Huang, "Pose-based temporal-spatial network (PTSN) for gait recognition with carrying and clothing variations," in *Proc. Chin. Conf. Biometric Recognit.*, 2017, pp. 474–483.
- [25] Y. Wen, K. Zhang, Z. Li, and Y. Qiao, "A discriminative feature learning approach for deep face recognition," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 499–515.
- [26] Y. Liu, H. Li, and X. Wang, "Rethinking feature discrimination and polymerization for large-scale recognition," 2017, *arXiv:1710.00870*. [Online]. Available: <https://arxiv.org/abs/1710.00870>
- [27] H. Wang *et al.*, "CosFace: Large margin cosine loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5265–5274.
- [28] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," 2018, *arXiv:1801.07698*. <https://arxiv.org/abs/1801.07698>
- [29] R. Zhao, W. Ouyang, and X. Wang, "Person re-identification by salience matching," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2528–2535.
- [30] D. Yi, Z. Lei, S. Liao, and S. Z. Li, "Deep metric learning for person re-identification," in *Proc. Int. Conf. Pattern Recognit.*, Aug. 2014, pp. 34–39.
- [31] Y. Sun, L. Zheng, Y. Yang, Q. Tian, and S. Wang, "Beyond part models: Person retrieval with refined part pooling (and a strong convolutional baseline)," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 480–496.
- [32] C. Song, Y. Huang, W. Ouyang, and L. Wang, "Mask-guided contrastive attention model for person re-identification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1179–1188.
- [33] P. Molchanov, X. Yang, S. Gupta, K. Kim, S. Tyree, and J. Kautz, "Online detection and classification of dynamic hand gestures with recurrent 3D convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 4207–4215.
- [34] Y. Du, W. Wang, and L. Wang, "Hierarchical recurrent neural network for skeleton based action recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1110–1118.
- [35] K. Simonyan and A. Zisserman, "Two-stream convolutional networks for action recognition in videos," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 568–576.
- [36] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, "Realtime multi-person 2D pose estimation using part affinity fields," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1302–1310.
- [37] R. A. Güler, N. Neverova, and I. Kokkinos, "DensePose: Dense human pose estimation in the wild," 2018, *arXiv:1802.00434*. [Online]. Available: <https://arxiv.org/abs/1802.00434>
- [38] Y. Makihara, R. Sagawa, Y. Mukaigawa, T. Echigo, and Y. Yagi, "Gait recognition using a view transformation model in the frequency domain," in *Proc. Eur. Conf. Comput. Vis.*, 2006, pp. 151–163.
- [39] W. Kusakunniran, Q. Wu, J. Zhang, H. Li, and L. Wang, "Recognizing gaits across views through correlated motion co-clustering," *IEEE Trans. Image Process.*, vol. 23, no. 2, pp. 696–709, Feb. 2013.
- [40] M. Hu, Y. Wang, Z. Zhang, J. J. Little, and D. Huang, "View-invariant discriminative projection for multi-view gait-based human identification," *IEEE Trans. Inf. Forensics Security*, vol. 8, no. 12, pp. 2034–2045, Dec. 2013.
- [41] M. Goffredo, I. Bouchrika, J. N. Carter, and M. S. Nixon, "Self-calibrating view-invariant gait biometrics," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 40, no. 4, pp. 997–1008, Aug. 2010.
- [42] W. Kusakunniran, Q. Wu, J. Zhang, Y. Ma, and H. Li, "A new view-invariant feature for cross-view gait recognition," *IEEE Trans. Inf. Forensics Security*, vol. 8, no. 10, pp. 1642–1653, Oct. 2013.
- [43] S. Yu *et al.*, "GaitGANv2: Invariant gait feature extraction using generative adversarial networks," *Pattern Recognit.*, vol. 87, pp. 179–189, Mar. 2019.
- [44] B. Hu, Y. Gao, Y. Guan, Y. Long, N. Lane, and T. Ploetz, "Robust cross-view gait identification with evidence: A discriminant gait gan (DiG-GAN) approach on 10000 people," 2018, *arXiv:1811.10493*. [Online]. Available: <https://arxiv.org/abs/1811.10493>
- [45] H. Chao, Y. He, J. Zhang, and J. Feng, "GaitSet: Regarding gait as a set for cross-view gait recognition," 2018, *arXiv:1811.06186*. [Online]. Available: <https://arxiv.org/abs/1811.06186>
- [46] J. Donahue *et al.*, "Long-term recurrent convolutional networks for visual recognition and description," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 2625–2634.
- [47] W. An, R. Liao, S. Yu, Y. Huang, and P. C. Yuen, "Improving gait recognition with 3D pose estimation," in *Proc. Chin. Conf. Biometric Recognit.* Cham, Switzerland: Springer, 2018, pp. 137–147.
- [48] G. Batchuluun, H. S. Yoon, J. K. Kang, and K. R. Park, "Gait-based human identification by combining shallow convolutional neural network-stacked long short-term memory and deep convolutional neural network," *IEEE Access*, vol. 6, pp. 63164–63186, 2018.
- [49] F. Battistone and A. Petrosino, "TGLSTM: A time based graph deep learning approach to gait recognition," *Pattern Recognit. Lett.*, to be published.
- [50] Z. Zhang *et al.*, "Gait recognition via disentangled representation learning," 2019, *arXiv:1904.04925*. [Online]. Available: <https://arxiv.org/abs/1904.04925>
- [51] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, "DeepFace: Closing the gap to human-level performance in face verification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1701–1708.

- [52] Y. Sun, Y. Chen, X. Wang, and X. Tang, "Deep learning face representation by joint identification-verification," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 1988–1996.
- [53] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 815–823.
- [54] W. Liu, Y. Wen, Z. Yu, and M. Yang, "Large-margin softmax loss for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 507–516.
- [55] D. Li, X. Chen, Z. Zhang, and K. Huang, "Learning deep context-aware features over body and latent parts for person re-identification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 7398–7407.
- [56] M. Jaderberg, K. Simonyan, A. Zisserman, and K. Kavukcuoglu, "Spatial transformer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 2017–2025.
- [57] H. Wang and L. Wang, "Modeling temporal dynamics and spatial configurations of actions using two-stream recurrent neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 499–508.
- [58] L. Pigou, A. Van Den Oord, S. Dieleman, M. Van Herreweghe, and J. Dambre, "Beyond temporal pooling: Recurrence and temporal convolutions for gesture recognition in video," *Int. J. Comput. Vis.*, vol. 126, nos. 2–4, pp. 430–439, 2015.
- [59] L. Chen, J. Lu, Z. Song, and J. Zhou, "Part-activated deep reinforcement learning for action prediction," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 421–436.
- [60] Y. Tang, Y. Tian, J. Lu, P. Li, and J. Zhou, "Deep progressive reinforcement learning for skeleton-based action recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5323–5332.
- [61] G. Pereyra, G. Tucker, J. Chorowski, L. Kaiser, and G. Hinton, "Regularizing neural networks by penalizing confident output distributions," 2017, *arXiv:1701.06548*. [Online]. Available: <https://arxiv.org/abs/1701.06548>
- [62] T. Chen *et al.*, "MXNet: A flexible and efficient machine learning library for heterogeneous distributed systems," 2015, *arXiv:1512.01274*. [Online]. Available: <https://arxiv.org/abs/1512.01274>
- [63] A. Borgia, Y. Hua, E. Kodirov, and N. Robertson, "Cross-view discriminative feature learning for person re-identification," *IEEE Trans. Image Process.*, vol. 27, no. 11, pp. 5338–5349, Nov. 2018.
- [64] L. Lee and W. E. L. Grimson, "Gait analysis for recognition and classification," in *Proc. 5th IEEE Int. Conf. Autom. Face Gesture Recognit.*, May 2002, pp. 155–162.
- [65] J. Liu, Y. Huang, L. Wang, and S. Wu, "Hierarchical feature coding for image classification," *Neurocomputing*, vol. 144, pp. 509–515, Nov. 2014.
- [66] Y. Liu, J. Yan, and W. Ouyang, "Quality aware network for set to set recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4694–4703.
- [67] S. Yu, D. Tan, and T. Tan, "A framework for evaluating the effect of view angle, clothing and carrying condition on gait recognition," in *Proc. Int. Conf. Pattern Recognit.*, 2006, pp. 441–444.
- [68] H. Iwama, M. Okumura, Y. Makihara, and Y. Yagi, "The OU-ISIR gait database comprising the large population dataset and performance evaluation of gait recognition," *IEEE Trans. Inf. Forensics Security*, vol. 7, no. 5, pp. 1511–1521, Oct. 2012.
- [69] N. Takemura, Y. Makihara, D. Muramatsu, T. Echigo, and Y. Yagi, "Multi-view large population gait dataset and its performance evaluation for cross-view gait recognition," *IPSI Trans. Comput. Vis. Appl.*, vol. 10, no. 1, pp. 1–14, 2018.
- [70] Y. Zhang, Y. Huang, L. Wang, and S. Yu, "A comprehensive study on gait biometrics using a joint CNN-based method," *Pattern Recognit.*, vol. 93, pp. 228–236, Sep. 2019.



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